

CONTROL OF ACTIVATED SLUDGE FILAMENTOUS BULKING—I. EFFECT OF THE HYDRAULIC REGIME OR DEGREE OF MIXING IN AN AERATION TANK

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Abstract—Laboratory experiments were conducted on four activated sludge systems with various flow patterns. The systems were characterized by dispersion numbers which are a numerical expression of the degree of mixing. Perfect mixing has a dispersion number of infinity. The dispersion numbers in the four systems were ∞ , 1.06, 0.17 and 0.033. All systems were operated at the same detention period of 8 h and approximately at the same sludge loadings. A mixture of starch and peptone was used as the substrate. The mixed cultures developed in the systems mentioned possessed average SVI values of 517, 300, 91 and 51 ml g⁻¹ respectively. The high SVI values were caused by a high content of the filamentous microorganisms, which consisted mainly of *Leucothrix* and *Sphaerotilus*, in the developed mixed cultures. It is concluded that the complete mixing systems tend to lead to excessive growth of the filamentous microorganisms. Aeration systems with a low degree of the axial mixing, i.e. with low dispersion numbers and higher concentration gradients of the substrate along the systems, lead to the growth of non-filamentous and suppress the growth of filamentous microorganisms. It is possible to control the growth of filamentous microorganisms in the mixed cultures by maintaining the concentration gradient of the substrate along the aeration system. The concentration gradient of the substrate depends on the degree of mixing in the aeration system.

INTRODUCTION

BULKING of activated sludge can be caused by numerous factors (PIPES, 1969). One frequent cause is excessive growth of filamentous microorganisms in a mixed culture. The causative microorganisms have been reported as e.g., *Sphaerotilus* (LACKEY and WATTIE, 1940; RUCHHOFF and KACHMAR, 1941; ADAMSE, 1968), *Geotrichum candidum* (JONES, 1964, 1965), *E. coli* (PASVEER, 1969; HÜNERBERG *et al.*, 1970), *Leucothrix cohaerens*, *Lineola longa*, and *Spirulina albiba* (CYRUS and SLADKA, 1970). High contents of filamentous microorganisms in activated sludge usually result in high values of sludge volume indices (SVI).

During the past decade, many experiments on continuous and semicontinuous cultivation of mixed cultures have been carried out in our laboratories. Continuous cultivation was mostly performed in completely mixed aeration tanks. The most striking phenomenon observed was the different composition of mixed cultures cultivated continuously and semicontinuously. Experiments with saccharides, fatty acids, amino acids, and phenols (CHUDOBA, 1967; CHUDOBA *et al.*, 1968a, 1968b) have shown that only slight amounts of filamentous microorganisms were present in mixed cultures cultivated semicontinuously, while mixed cultures cultivated continuously in complete mixing aeration tanks were composed mostly of filamentous microorganisms. Experiments with starch (GRAU and DOHANYOS, 1971) have shown that mixed cultures cultivated semicontinuously had values of SVI in the range from 50 to 250 ml g⁻¹, while those cultivated continuously in complete mixing aeration tanks had SVI in the range from 450 to 850 ml g⁻¹. Mixed cultures developed on

Prague settled sewage (KOLLER, 1966) in semicontinuous systems with detention periods of 3, 6 and 12 h possessed average values of SVI of 51, 72 and 58 ml g⁻¹, respectively, while those cultivated in continuous complete mixing systems at detention periods of 2.4, 3.7 and 6.4 h possessed average values of SVI of 280, 512 and 440 ml g⁻¹, respectively.

These experiments indicated that the method of cultivation might have some influence on the selection of microorganisms in a mixed culture. On the other hand, however, this phenomenon might be attributed to the influence of the type of a reactor used (complete mixing or piston flow). A semicontinuous reactor is usually used to simulate conditions which would exist in a continuous reactor with piston flow. The latter ideal system is not practically realizable.

Experiments carried out in the British Water Pollution Research Laboratory (1969) indicated that the flow pattern influenced the settling characteristics of the activated sludge. A series of six small activated-sludge plants differing in the number of aeration compartments connected in series was brought into operation over a period of 31 days. The operating conditions were (the first numbers for plants No. 1, 3, 5, the second ones for plants No. 2, 4, 6): detention periods 6 and 12 h; MLSS concentrations 5200–5400 and 3500–4100 mg l⁻¹; sludge loads 0.19–0.20 and 0.12–0.13 (kg.day)⁻¹. The mean BOD₅ of the settled effluents was 4.2–5.0 and 2.1–4.5 mg l⁻¹. The settling properties and the density of sludges differed as indicated by the values of SVI:

Plant No.	1	2	3	4	5	6
No. of compartments	1	1	2	2	4	4
SVI (ml g ⁻¹)	46–162	50–203	44–129	55–125	42–73	49–75

The conclusion of this report was: "It would appear that in the case of plants operated at loadings less than 0.2 g BOD g⁻¹ sludge per day it is beneficial to have a number of compartments in series".

The conditions at higher loadings are, however, of interest, and no quantitative relationship between the sludge behavior and the hydraulic regime in an aeration tank has been studied yet. The object of this study was to find if any relation exists between the hydraulic regime in an aeration tank and the composition of microorganisms in a mixed culture.

Both ideal systems—with complete mixing and piston flow—represent two extremes. Between them there exists a continuous spectrum of systems with non-ideal flow. Continuous flow systems can be characterized by a dispersion number, D/uL , which has the value of zero for piston flow and of infinite for complete mixing (LEVENSPIEL, 1962). Continuous systems with plug flow pattern can be modelled by a tank-in-series model (LEVENSPIEL, 1962) or by compartmentalization of a reaction tank (THOMAS and MCKEE, 1944; MILBURY *et al.*, 1965).

DESCRIPTION OF EXPERIMENTS

Four laboratory continuous flow models were operated. All were operated under similar conditions which are summarized in TABLE 1. The air used for aeration was purified in an oil separator and filtered through cotton-wool and activated carbon filters. It was supplied in excess as indicated by the concentration of dissolved oxygen, which never fell below 2 mg l⁻¹ anywhere in the aeration vessels. Influent was dosed into aeration tanks by electrolysis pumps described by SYMONS (1963). Glass settling

TABLE 1. AVERAGE VALUES OF OPERATING PARAMETERS OF INDIVIDUAL SYSTEMS AND SLUDGE CHARACTERISTICS

Parameters	Units	System			
		I $D/uL = \infty$	II $D/uL = 1.06$	III $D/uL = 0.17$	IV $D/uL = 0.033$
Duration of experiments	day	70	30	80	70
Averages from a period	day	35-65	20-30	47-77	35-65
Total volume of the aerated mixture	l.	4	4	4	4
Detention period	h	8	8	8	8
Recirculation ratio	—	1-3	0.8-1.2	0.8-1.2	0.8-1.2
Loading	kg BOD ₅ m ⁻³ day ⁻¹	0.84	0.84	0.84	0.84
	kg COD m ⁻³ day ⁻¹	1.95	1.95	1.95	1.95
	kg BOD ₅ (kg MLVSS.day) ⁻¹	0.48	0.28	0.31	0.30
	kg COD (kg MLVSS.day) ⁻¹	1.11	0.65	0.73	0.69
MLSS	g l ⁻¹	1.9	3.28	2.94	3.1
MLVSS	%	92.6	91.5	91.4	93.2
Sludge age	day	3.0	4.8	4.7	4.5
SVI	ml g ⁻¹	517	300	91	51
COD of sludge	g g ⁻¹ VSS	1.40	1.25	1.43	1.42
BOD ₅ of sludge	g g ⁻¹ VSS	0.62	—	0.70	0.59
Respiration rate at outlet from the aeration system	mg(g VSS.h) ⁻¹	14.0	9.4	13.9	16.2
Content of N in MLVSS	%	10.1	13.9	13.0	10.0

tanks connected with the aeration tanks were 8 cm in diameter and their volume was 1.5 l. Activated sludge was recirculated by means of an air lift. All systems were started with activated sludge from the Prague sewage treatment plant. Experiments were carried out in two periods. In the first (from October to December 1971), systems with $D/uL = \infty$ and $D/uL = 0.033$ were operated, in the second (from February to April 1972), systems with $D/uL = 1.06$ and $D/uL = 0.17$ were operated. All experiments were conducted at room temperature varying from 15 to 20°C.

Description of laboratory models

System I. A schematic outline of this system is shown in FIG. 1. A glass cylindrical vessel, 17.5 cm dia., was used as the aeration tank. The hydraulic characteristics, which are given with the Results, indicated that this was the ideal complete mixing reactor with $D/uL = \infty$.

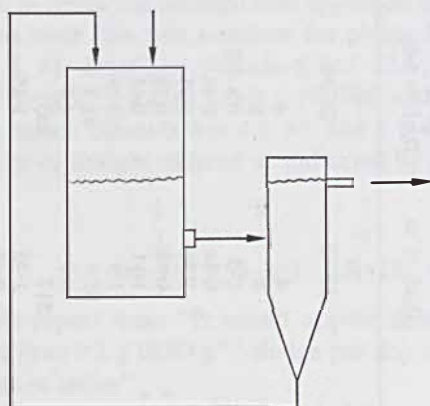


FIG. 1. The schematic outline of system I.

System II (FIG. 2). A rectangular plexiglass tank, 17 × 17 × 25 cm, was divided by walls into four compartments each 1-l. in volume. The connection between the individual compartments was through holes situated at the bottom in the walls. The diameter of the holes was 2.2 cm. The dispersion number of the system was 1.06.

System III (FIG. 3). A rectangular plexiglass tank, 26 × 13 × 25 cm, was divided by walls into eight compartments each 0.5 l. in volume. The connection between the

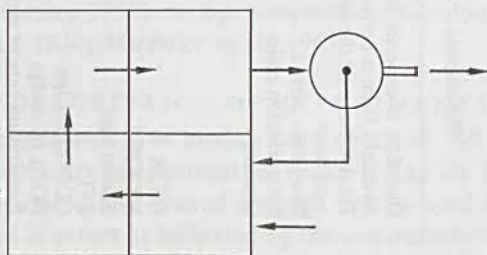


FIG. 2. The schematic outline of system II.

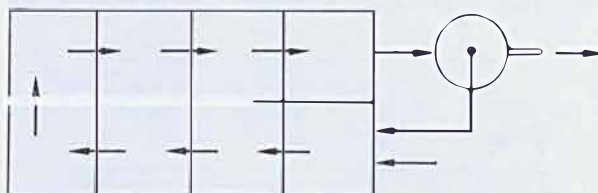


FIG. 3. The schematic outline of system III.

individual compartments was similar to that in system II. The diameter of the holes was 1.0 cm. The dispersion number of the system was 0.17. The overall view of the system in operation is shown in FIG. 4.

System IV. A schematic outline of this system is shown in FIG. 5. It consisted of 16 glass tubes in series connected at the bottom (FIG. 6). The volume of each compartment was 0.25 l. The dispersion number of the system was 0.033.

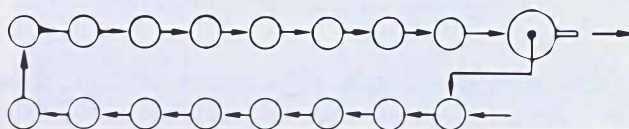


FIG. 5. The schematic outline of system IV.

Composition of the substrate used

The same substrate was used in all systems during both periods and is given in TABLE 2. A slight difference in the organic nitrogen content was probably due to different batches of peptone.

Treatment of samples before analysis

A "shaken sample" was prepared from collected effluents by shaking. In this sample, only suspended solids were determined. A "settled sample" was prepared from the "shaken sample" by sedimentation in 2-l. cylinders for a period of 0.5 h. Supernatants were siphoned off and analysed. A "filtered sample" was prepared from the "settled sample" by filtering through a membrane filter with an average pore size of $0.6 \mu\text{m}$ (for all determinations except of BOD_5) or through a candle filter with an average pore size of $1 \mu\text{m}$ (for BOD_5 determinations).

Determination of dispersion numbers of the aeration systems

In order to characterize the systems hydraulically, tracing tests were conducted using Rhodamine B as a tracer. This tracer was used, e.g. by MURPHY and TIMPANY (1967) in laboratory experiments and by MURPHY and BOYKO (1970) in a full scale aeration tank. In these tests, the aeration systems were filled with tap water and recirculation from the settling tanks was stopped. However, the influents were doubled in order to keep the same theoretical detention periods as with the recirculation ratio of 1. This value had been planned as the average for further experiments. Samples for determination of the tracer were pipetted off at the outlet from the last

TABLE 2. AVERAGE COMPOSITION OF INFLUENTS IN BOTH PERIODS

Item	First period (mg l ⁻¹)	Second period (mg l ⁻¹)
Peptone	300	300
Starch	300	300
NaHCO ₃	80	80
KH ₂ PO ₄	8	8
NH ₄ ⁺	3.2	3.4
NO ₃ ⁻	4.2	4.0
Organic N	42.0	43.0
Total N	45.0	46.5
PO ₄ ³⁻	10.5	10.8
P	3.5	3.6
COD	650	650
BOD ₅	280	280

(All components were dissolved in tap water)

compartment or tube of the given system. Thus, though settling tanks were connected with the aeration systems during these tests, they were not included into the systems. Sampling periods were 5, 10 or 15 min. Dispersion numbers were calculated by the method of successive approximations from the following equation (VAN DER LAAN, 1958):

$$\sigma^2 = 2 \frac{D}{uL} - 2 \left(\frac{D}{uL} \right)^2 \left[1 - \exp \left(- \frac{uL}{D} \right) \right]$$

where σ^2 is the determined value of variance.

ANALYTICAL METHODS

COD was determined by the dichromate method with the silver catalyst (*Standard Methods*, 1971). The method using 0.05 N solutions of potassium dichromate and ferrous ammonium sulphate was applied to samples with expected COD values below 100 mg l⁻¹. Corrections were provided for the nitrite content.

BOD₅ was determined by the dilution method (*Standard Methods*, 1971). The method with 0.10 M ammonium chloride was used for the elimination of nitrification (SIDDIQI *et al.*, 1967).

MLSS concentrations were determined by the membrane filtration method using membrane filters with average pore size of 1.5 μ m.

Suspended solids in effluents were estimated according to the following formulae:

$$SS_{\text{shaken sample}} = \frac{\text{COD}_{\text{shaken sample}} - \text{COD}_{\text{filtered sample}}}{\text{COD of MLSS}}$$

$$SS_{\text{settled sample}} = \frac{\text{COD}_{\text{settled sample}} - \text{COD}_{\text{filtered sample}}}{\text{COD of MLSS}}$$

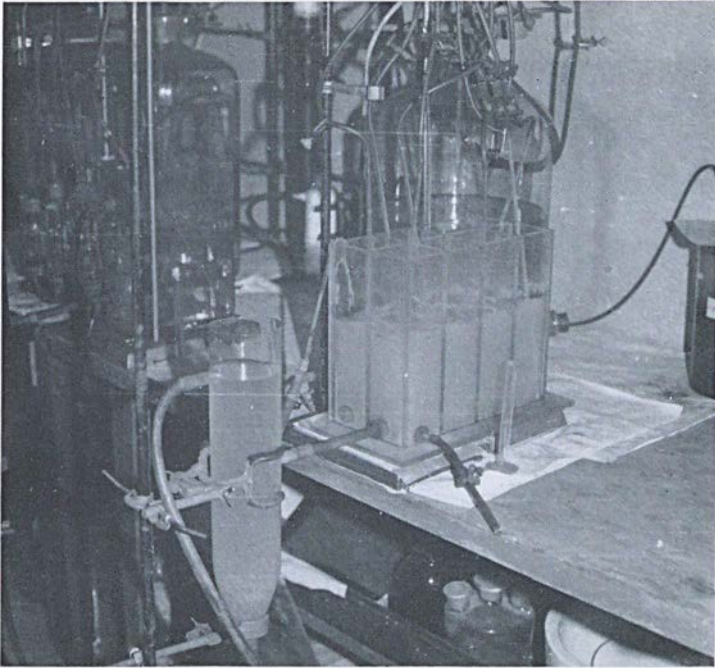


FIG. 4. The overall view of system III at operation.

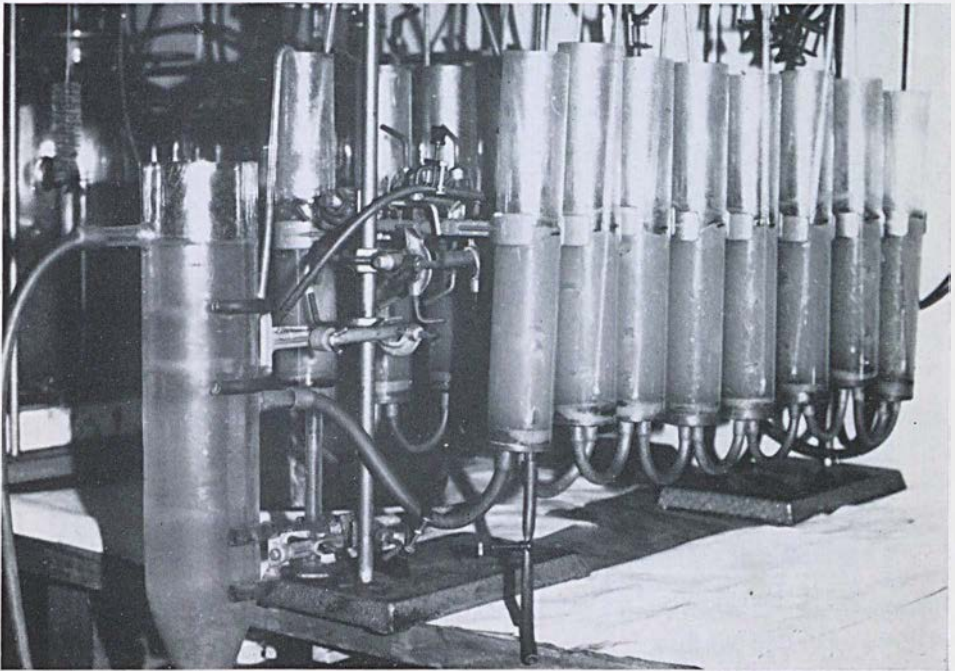


FIG. 6. The view of system IV at operation.

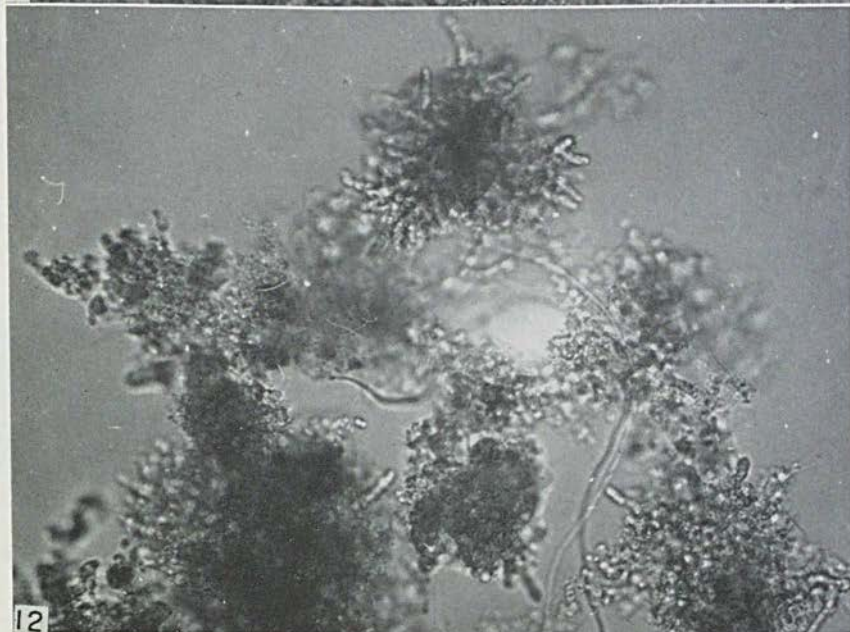


FIG. 11. The structure of the mixed culture developed in the complete mixing system.

FIG. 12. The structure of the mixed culture developed in system IV.

Respiration rates were measured by means of a membrane oxygen electrode at 20°C.

Volumes of activated sludge after 0.5 h of settling were mostly determined in 100 ml calibrated cylinders. One-litre cylinders were used to determine the relationship between SVI and suspended solids concentration.

Nitrogenous compounds, organic nitrogen, and phosphates were determined according to the Czechoslovak *Standard Methods* (1965).

RESULTS

Hydraulic characteristics of the systems

The hydraulic characteristic of the system I is shown in FIG. 7. The relationship between $-\log C/C_0$ and $t/\bar{t}$ should be linear for the complete mixing system with the slope of $p = -\log e = -0.434$ and the intercept of $q = 0$. Since the relationship in FIG. 7 has $p = -0.4$ and $q = 0$, it is evident that the system I can be taken for the deal complete mixing system.

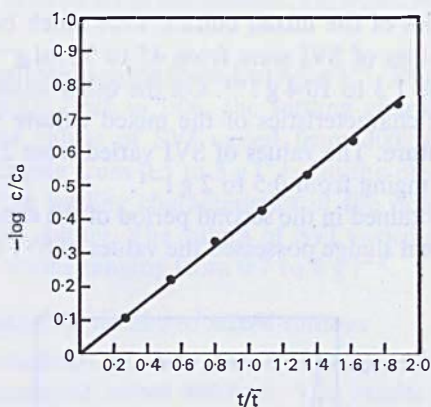


FIG. 7. Hydraulic characteristic of system I. $D/ul = \infty$, $\bar{t} = 3.74$ h.

The hydraulic characteristics of the systems II, III and IV are shown in FIG. 8. The dispersion numbers calculated were, respectively, 1.06, 0.17 and 0.033.

Settling characteristics of the developed mixed cultures

Settling characteristics of the activated sludge may be evaluated by the sludge volume index. The average values of SVI are given in TABLE 1 and were, respectively, 517, 300, 91 and 51 ml g⁻¹ with the mixed cultures developed in the systems I, II, III and IV. These are surprising differences.

During the experiments, measurements were conducted on the dependence of SVI on the MLSS concentrations. Results obtained in the first period of the experiments are shown in FIG. 9. The original sludge from the Prague sewage treatment plant possessed the values of SVI from 170 to 250 ml g⁻¹ with the MLSS concentrations ranging from 0.8 to 3.8 g l⁻¹. It can be seen in FIG. 9 that settling characteristics of the original sludge changed markedly and differentially during the further cultivation both in the system I and in the system IV. In the system IV, with $D/ul = 0.033$,

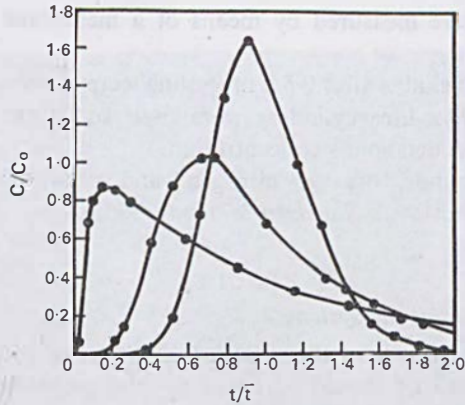


FIG. 8. Hydraulic characteristic of systems II, III and IV. System II: $D/uL = 1.06$, $\bar{t} = 3.48$ h (left curve); System III: $D/uL = 0.17$, $\bar{t} = 3.79$ h (middle curve); System IV: $D/uL = 0.033$, $\bar{t} = 3.85$ h (right curve).

the settling characteristics of the mixed culture were much better than those of the original culture. The values of SVI were from 45 to 75 ml g⁻¹ with the MLSS concentrations ranging from 1.3 to 10.4 g l⁻¹. On the other hand, in the system I, with $D/uL = \infty$, the settling characteristics of the mixed culture were much worse than those of the original culture. The values of SVI varied from 250 to 1000 ml g⁻¹ with MLSS concentrations ranging from 0.5 to 2 g l⁻¹.

Similar results were obtained in the second period of the experiments. It can be seen in FIG. 10 that the original sludge possessed the values of SVI from 115 to 190 ml g⁻¹

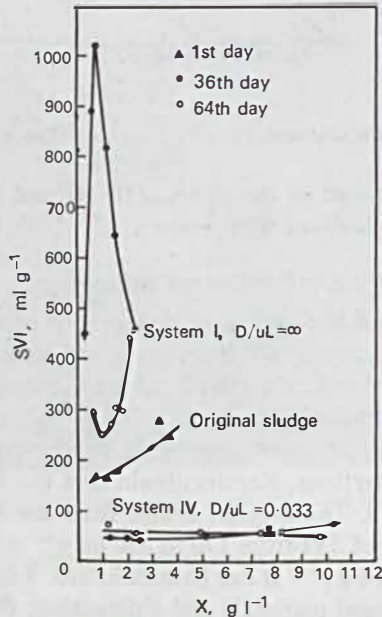


FIG. 9. Dependence of SVI on the MLSS concentration. The first period of the experiments.

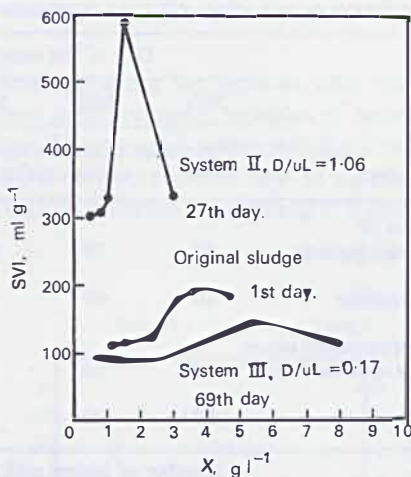


FIG. 10. Dependence of SVI on the MLSS concentration. The second period of the experiments.

with the MLSS concentrations ranging from 1 to 4.7 g l⁻¹. During the further cultivation in the system II, with $D/uL = 1.06$, the settling characteristics of the mixed culture deteriorated. The values of SVI varied from 300 to 590 ml g⁻¹ with the MLSS concentrations ranging from 0.5 to 3 g l⁻¹. On the other hand, in the system III, with $D/uL = 0.17$, the settling characteristics of the mixed culture were better than those of the original culture. The values of SVI varied from 90 to 145 ml g⁻¹ with the MLSS concentrations ranging from 0.7 to 8 g l⁻¹.

Microscopic examinations of the developed mixed cultures

Regular microscopic observations were carried out to characterize the composition and structure of the developed mixed cultures. The results of these observations obtained in the first period of the experiments are summarized in TABLES 3 and 4. The microscopic examinations revealed the causes of the different settling characteristics of the mixed cultures developed in the systems I and IV. While the square representation of the filamentous microorganisms in the microscopical field was 70–90 per cent in the mixed culture from the complete mixing system, it amounted to 15–30 per cent in the system IV. It is of interest that a great part of the filamentous microorganisms consisted of *Cyanophyta apochlorotica*, namely *Leucothrix*. It formed about 75 per cent of the filamentous microorganisms in the complete mixing system and about 30–70 per cent in the system IV. The rest of the filamentous microorganisms consisted mostly of *Sphaerotilus*. The different structures of the mixed cultures developed in the systems I and IV can be seen in FIGS. 11 and 12, respectively.

In the second period, the square representations of the filamentous microorganisms in the mixed cultures developed in the systems II and III were approximately 70 and 50 per cent, respectively. Again, *Leucothrix* formed about 70 and 20–50 per cent of the filamentous microorganisms in the systems II and III, respectively.

Thus, it is evident, that the different settling characteristics of the mixed cultures developed in hydraulically different systems, expressed by means of SVI, were caused by various contents of the filamentous microorganisms in the mixed cultures.

TABLE 3. COMPOSITION OF THE MIXED CULTURE DEVELOPED IN SYSTEM I

Microorganisms	Day of the experiments			
	39th	46th	53rd	65th
	Percentage of square representation in a microscopical field			
Total representation of filamentous microorganisms	80	70	90	80
<i>Cyanophyta apochlorotica</i>	60	60	70	60
Filamentous microorganisms other than <i>Cyanophyta apochlorotica</i>	20	10	20	20
<i>Zoogloea</i>	20	30	10	20
Number of higher microorganisms per 1 g^{-1} of MLSS				
<i>Flagellata apochrom.</i>	1×10^6	1.7×10^6	0.6×10^6	1.2×10^6
<i>Ciliata</i> { free	3.8×10^5	1.1×10^8	1.2×10^6	4.1×10^6
{ fixed	4.7×10^6	2.8×10^6	5.8×10^6	5.8×10^5
<i>Rhizopoda</i>	3.3×10^7	1.7×10^7	0.6×10^6	—
<i>Vermes</i> { <i>Rotatoria</i>	—	—	—	2.9×10^5
{ <i>Nematoda</i>	—	—	2.9×10^5	2.3×10^4

TABLE 4. COMPOSITION OF THE MIXED CULTURE DEVELOPED IN SYSTEM IV

Microorganisms	Day of the experiments			
	39th	46th	53rd	65th
	Percentage of square representation in a microscopical field			
Total representation of filamentous microorganisms	30	20	15	30
<i>Cyanophyta apochlorotica</i>	10	10	10	10
Filamentous microorganisms other than <i>Cyanophyta apochlorotica</i>	20	10	5	20
<i>Zoogloea</i>	70	80	85	70
Number of higher microorganisms per 1 g^{-1} of MLSS				
<i>Flagellata apochrom.</i>	4.0×10^5	5.6×10^5	6.3×10^6	4.1×10^4
<i>Ciliata</i> { free	2.0×10^6	2.8×10^5	4.7×10^6	—
{ fixed	8.0×10^6	1.4×10^6	3.2×10^4	8.3×10^6
<i>Rhizopoda</i>	1.2×10^7	2.2×10^7	2.5×10^7	3.3×10^6
<i>Vermes</i> { <i>Rotatoria</i>	2.0×10^6	8.4×10^3	—	3.7×10^6
{ <i>Nematoda</i>	4.0×10^4	5.6×10^3	—	2.9×10^4

Changes in substrate concentrations and respiration rates along the individual systems

The substrate concentrations along the systems were measured and expressed as COD in solution. For these measurements, samples of aerated mixtures were filtered through a dense paper filter. The system I, which was the ideal complete mixing system, had uniform values of COD and respiration rate in all places of the aeration tank. The average value of COD in solution was 40 mg l^{-1} and that of respiration rate was $14 \text{ mg (g VSS.h)}^{-1}$.

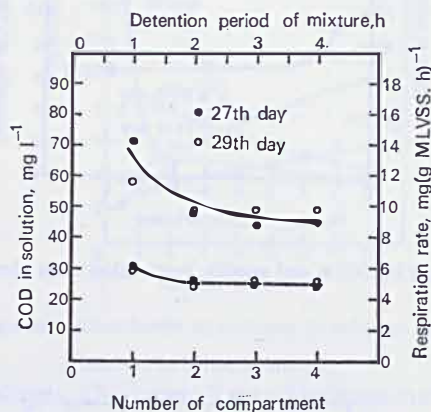


FIG. 13. Changes in CODs and specific respiration rates along system II.

Changes in the COD and specific respiration rate along the systems II, III and IV are shown in Figs. 13–15. The lowest COD gradient, from 30 to 25 mg l^{-1} , was measured in the system II, the highest one, from 105–135 to 40 mg l^{-1} , was measured in the system IV. The COD gradient in the system III was from 45 to 32 mg l^{-1} . Similar changes were found with the specific respiration rates. The decreases in the respiration rates were from 12–14 to 9–10 mg (g VSS.h)^{-1} along the system II, from 20–24 to $14 \text{ mg (g VSS.h)}^{-1}$ along the system III, and from 34–42 to 15–18 mg (g VSS.h)^{-1} along the system IV. It is of interest to notice in Figs. 14 and 15 that although the exogenous substrate was completely removed in the first quarter of the aeration system and COD in solution remained constant throughout the further parts of the system, the respiration rates decreased gradually up to the last compartment or tube. The decrease in the respiration rates was much slower along the second half of the system than along the first one.

In the course of the experiments, measurements were also conducted on the changes in various nitrogenous compounds along the systems. As an example, FIG. 16 is presented to show these changes along the system IV, in which they were highest. It is of interest to notice that nitrification proceeded from the inlet end of the aeration system in spite of the high substrate concentrations and loads in those first parts of the system. A slight increase in total soluble nitrogen throughout the second half of the system was probably due to the release of some nitrogenous compounds from the microorganisms. This phenomenon was not regularly observed at other measurements.

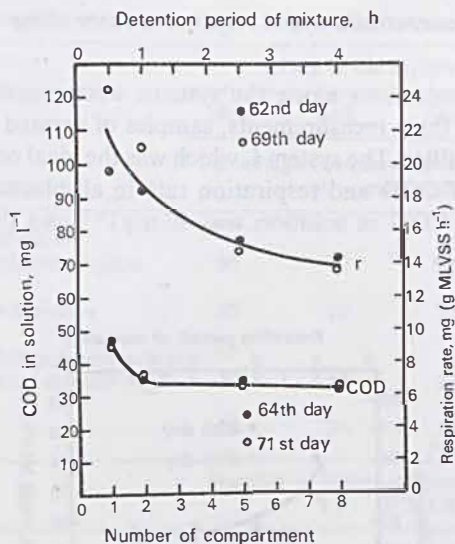


FIG. 14. Changes in CODs and specific respiration rates along system III.

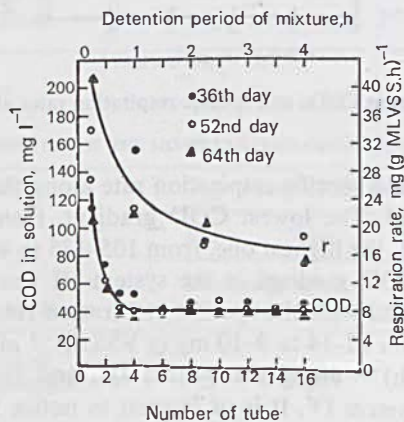


FIG. 15. Changes in CODs and specific respiration rates along system IV.

Purification efficiencies of the individual systems

Average data on suspended solids, COD and BOD₅ values in both settled and filtered effluents are summarized in TABLE 5. It can be seen that COD and BOD₅ values in filtered effluents were approximately the same with all systems. The lower COD values in the effluents from the systems II and III were probably due to different charges of peptone used in the second period of the experiments. The BOD₅:COD ratios in the filtered samples indicate that residual soluble organic matter was biologically resistant and nondegradable. We have evidence that it consists partially of resistant compounds originally present in peptone and partially of resistant waste

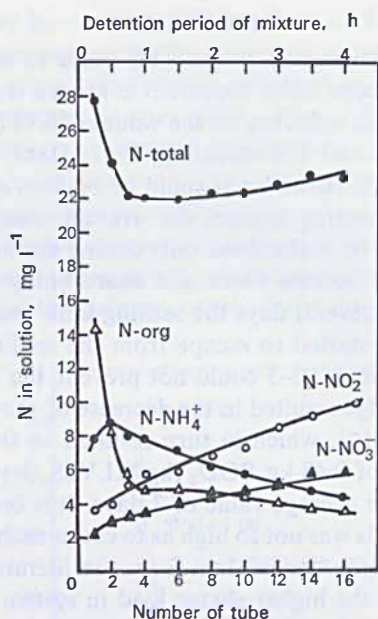


FIG. 16. Changes in various forms of nitrogen in solution along system IV.

products of the mixed cultures. The data in TABLE 5 indicate that most of the removable organic pollution remaining in the settled effluents was due to the suspended solids. Their concentrations were highest in the effluents from the complete mixing system and lowest in the effluents from the system II with $D/\mu L = 1.06$.

TABLE 5. AVERAGE VALUES OF SUSPENDED SOLIDS, COD, BOD₅, NITROGENOUS COMPOUNDS AND PHOSPHATES IN EFFLUENTS

Items	Units	System			
		I $D/\mu L = \infty$	II $D/\mu L = 1.06$	III $D/\mu L = 0.17$	IV $D/\mu L = 0.033$
Suspended solids					
shaken sample	mg l ⁻¹	86.0	12.1	15.2	24.0
settled sample	mg l ⁻¹	53.0	8.9	13.1	15.0
COD					
settled sample	mg l ⁻¹	107.8	30.6	43.0	50.7
filtered sample	mg l ⁻¹	33.2	19.4	25.0	31.3
BOD ₅					
settled sample	mg l ⁻¹	40.8	11.0	10.2	10.8
filtered sample	mg l ⁻¹	1.8	0.8	0.8	0.8
BOD ₅ :COD					
settled sample	mg l ⁻¹	0.38	0.36	0.24	0.14
filtered sample	mg l ⁻¹	0.05	0.04	0.03	0.03
NH ₄ ⁺	mg l ⁻¹	2.9	0.4	3.4	3.6
NO ₂ ⁻	mg l ⁻¹	29.0	7.9	17.9	22.0
NO ₃ ⁻	mg l ⁻¹	16.3	33.5	23.1	25.9
N-organic	mg l ⁻¹	4.9	4.0	3.9	4.9
N-total	mg l ⁻¹	19.6	14.3	17.2	20.3
PO ₄ ³⁻	mg l ⁻¹	3.1	0.9	0.2	2.0

DISCUSSION

In the course of the experiments, efforts were made to operate all the systems at approximately the same sludge loads because it is known from the literature that the sludge load might have some influence on the values of SVI (LOGAN and BUDD, 1956; ORFORD *et al.*, 1961; FORD and ECKENFELDER, 1967; DART and SPURR, 1968; BOON, 1970; GANCZARZYK, 1970). However, it could be realized only in the systems II, III and IV. In the complete mixing system, the average sludge load of $0.3 \text{ kg BOD}_5 (\text{kg MLVSS} \cdot \text{day})^{-1}$ could be maintained only during the first 15 days of operation. During this period sludge became more and more bulky and accumulated in the settling tank. In following several days the settling tank was full of sludge. Then the mostly filamentous sludge started to escape from the settling tank. Even increasing the recirculation ratio from 1 : 2–3 could not prevent the sludge from periodically escaping. The losses of sludge resulted in the decrease of the MLSS concentration to the average value of 1.9 g l^{-1} , which in turn resulted in the increase of the sludge load to the average value of $0.48 \text{ kg BOD}_5 (\text{kg MLVSS} \cdot \text{day})^{-1}$. Simultaneously, the sludge age decreased to the average value of 3 days. It is believed, however, that the difference in the sludge loads was not so high as to cause such enormous differences in the values of SVI. The examination of data from the literature cited above confirms this. It seems evident that the higher sludge load in system I did not cause the high SVIs, but, on the contrary, the high SVIs caused escaping sludge from the settling tank and in turn increasing the sludge load.

Experiments with the complete mixing system were repeated twice, with new original sludges and the systems were operated for the periods of three months. In both cases the results were similar. With the substrate used, the complete mixing system, operated at the average parameters given above, gave rise to excessive growth of filamentous microorganisms. With all experiments, the settling characteristics of the new mixed cultures were much worse than those of the original sludge.

The experiments described herein have indicated that a relation might exist between SVI and the dispersion number of the aeration system. This relationship is shown in FIG. 17 in which SVI is plotted versus the reciprocal of the dispersion number. The

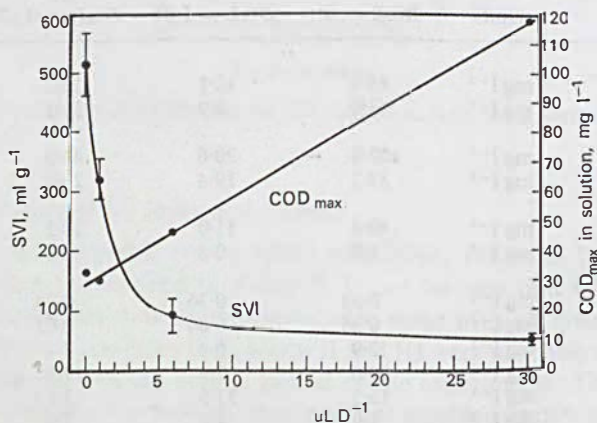


FIG. 17. Relationships between SVI, COD_{max} in solution and the reciprocal of the dispersion number. (Values of SVI refer to the MLSS concentrations in the range from 2 to 3 g l^{-1} .)

microscopic examinations have revealed that values of SVI were dependent on the content of filamentous microorganisms in the mixed cultures. It means that a degree of mixing, characterized quantitatively by the dispersion number, has an influence on the selection of microorganisms in the mixed culture. The degree of mixing in an aeration system determines a concentration gradient of the substrate along the system or a maximum value of the substrate which would be maintained at the inlet end of the system. The dependence of the maximum values of soluble COD on the reciprocal of the dispersion number is also shown in FIG. 17. With the exception of the system I, the values of COD_{max} are those measured in the first compartments or tube of the given system. It is felt that the primary cause of the selection of microorganisms in

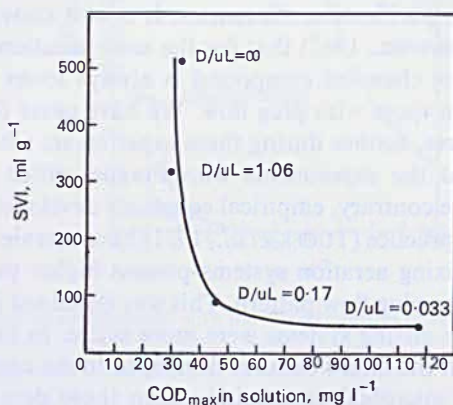


FIG. 18. Relationship between SVI and COD_{max} in solution. (Values of SVI refer to the MLSS concentrations in the range from 2 to $3\ g\ l^{-1}$.)

the mixed cultures is an actual concentration of the substrate at the inlet end of a given system. Its value is, of course, dependent on the degree of mixing. This is supported by experimental data not presented herein. They will be published in the second part of this series. The relationship between SVI and COD_{max} is shown in FIG. 18.

The experiments have revealed that the composition of the mixed cultures continuously change with the degree of dispersion. The system with the highest degree of dispersion, i.e. the complete mixing one gave rise to excessive growth of the filamentous microorganisms. However, we do not assert that all complete mixing systems necessarily produce mixed cultures with high contents of the filamentous microorganisms. There are other factors, e.g. sludge load and age, the composition of the substrate etc., which may influence the microbial composition of the mixed cultures. For example, a mixed culture developed on the same substrate described above in the complete mixing system operated at sludge load of $0.03\ kg\ (kg\ MLVSS \cdot day)^{-1}$, detention period of 48 h, and sludge age of 120 days possessed the values of SVI in the range from 60 to $200\ ml\ g^{-1}$.

The systems with the low degree of dispersion and with the higher concentration gradients of the substrate along the first parts of the systems produced growths of zoogloal form and suppressed the growth of the filamentous microorganisms. It seems that in practice these systems would be advantageous for the treatment of such waste

waters which are known to cause the filamentous bulking. As can be seen from FIGS. 14 and 15, the tapered aeration is necessary for these systems. However, they are not recommended for the treatment of such waste waters which contain higher concentrations of toxic compounds (e.g. phenols, formaldehyde, etc.).

The results presented in FIGS. 9 and 10 have confirmed the facts known from the literature (ISENBERG and HEUKELEKIAN, 1959; SOMERS, 1968; DICK and VESILIND, 1969) that values of SVI are dependent on the MLSS concentration. It can be seen in FIGS. 9 and 10 that this dependence is more expressive with bulking sludges than with those with good settling characteristics. Similar conclusions can be drawn from the data of SOMERS (1968).

At the given loading parameters and detention periods all systems possessed approximately the same purification efficiencies. It is well known from the theory of chemical reactors (LEVENSPIEL, 1962) that for the same reaction period the degree of conversion of a reacting chemical compound is always lower in reactors with ideal complete mixing than in those with plug flow. We have never found this in aeration tanks with mixed cultures, neither during these experiments with starch and peptone as the substrate nor at the experiments with Prague settled sewage (TUČEK and CHUDOBA, 1969). On the contrary, empirical equations developed on the basis of data from sewage treatment practice (TUČEK *et al.*, 1971) have revealed that under the same conditions complete mixing aeration systems possess higher purification efficiencies than the systems with the plug flow pattern. This was explained by the fact that mixed cultures in the complete mixing systems were more active. In the light of the present findings it is evident that the mixed cultures developed in the complete mixing systems might have a different microbial composition than those developed in the systems with plug flow pattern. As has been shown, the filamentous microorganisms prevail in the mixed cultures developed in the complete mixing systems. We have experimental evidence that they are more active towards the substrate used. It is in good agreement with the findings of TISCHLER and ECKENFELDER (1969) who measured higher specific rates of glucose and phenol removal with filamentous than non-filamentous cultures.

It means that required purification efficiencies can be obtained in the complete mixing systems with shorter detention periods. This would result in smaller aeration volumes and in turn in lower construction costs. However, the filamentous mixed cultures developed in the complete mixing systems have poor settling characteristics and would demand large secondary settling tanks with large squares and volumes. This can be demonstrated by the solids flux curves (DICK, 1970) which are shown in FIG. 19 for original sludge and those developed in the systems I and IV. Thus, it is evident that the construction costs saved on the aeration tank should have to be expended on the secondary settling tank.

The results summarized in TABLE 5 have confirmed the facts known from sewage treatment practice (HASELTINE, 1932; HEUKELEKIAN, 1941; KEEFER, 1963) that bulked sludges usually produce effluents of better quality than non-bulked ones. This phenomenon can be attributed to the concentration of suspended solids in settled effluents. In our experiments, the lowest concentrations of suspended solids in the settled effluents were measured in the system II with the average value of SVI 300 ml g^{-1} . KEEFER (1963) has measured the lowest concentrations of suspended solids in effluents at the sludge volume indices from 220 to 260 ml g^{-1} . The system IV produced mixed cultures with very good settling characteristics, but effluents con-

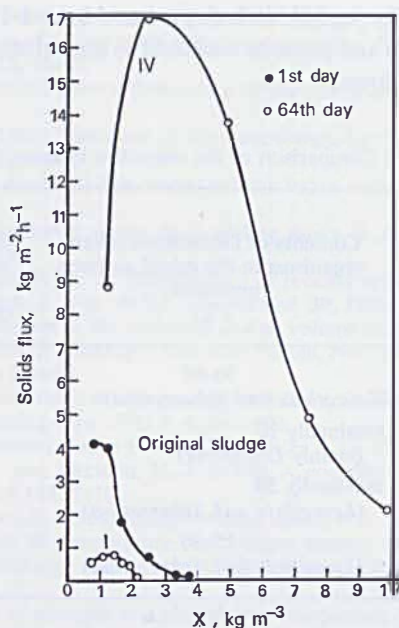


FIG. 19. The solids flux curves obtained with the mixed cultures developed in systems I and IV.

tained very small particles reminding pinpoint flocs (PIPES, 1969). These small flocs were not present in the effluents from the systems with higher contents of the filamentous microorganisms. The filamentous structure acts probably as a net in which small particles are entrapped.

It is felt that systems with both too high and too low SVIs are not desirable in practice. The former ones produce very clear effluents, but mixed cultures can be thickened and handled with difficulty. Mixed cultures in the later ones can be thickened and handled with ease, but the effluents contain usually higher concentrations of fine suspended solids. It seems that an optimal system should exist that would produce acceptable values both of SVI and suspended solids in effluents. On the basis of our experiments, the system III with $D/uL = 0.17$ seems to be the optimal one. The average concentration of the suspended solids in the settled effluents was 13 mg l^{-1} and SVI had the acceptable value of 90 ml g^{-1} . It is possible that even from the systems with very low SVIs very clear effluents could be obtained by means of proper controlling solids retention time. WALKER (1971) has presented experimental evidence that this is best realizable hydraulically.

SUMMARY

Laboratory experiments were conducted on four activated sludge systems providing different degrees of mixing. The systems were characterized by dispersion numbers which numerically express the degree of mixing and all were operated at the same detention period of 8 h and at the same volume loadings. The sludge loadings were

between 0.28–0.48 kg BOD₅ (kg MLVSS .day)⁻¹ and 0.65–1.11 kg COD (kg MLVSS .day)⁻¹. A mixture of starch and peptone was used as the substrate. The results obtained may be summarized as follows:

(1) Comparison of the respective systems:

System:	Dispersion number of the system:	Contents of filamentous micro-organisms in the mixed cultures produced: (%)	Average value of SVI: (ml g ⁻¹)	Concentration gradient of COD along the system: (mg l ⁻¹)
I	$D/ul = \infty$	70–90 (<i>Leucothrix</i> and <i>Sphaerotilus</i>)	517	0
II	$D/ul = 1.06$	maximally 70 (mainly <i>Leucothrix</i>)	300	from 20 to 25
III	$D/ul = 0.17$	maximally 50 (<i>Leucothrix</i> and <i>Sphaerotilus</i>)	91	from 45 to 32
IV	$D/ul = 0.033$	15–30 (<i>Leucothrix</i> and <i>Sphaerotilus</i>)	51	from 105–135 to 40

(2) The various values of SVI were related to content of filamentous microorganisms in the mixed cultures developed in the individual aeration systems.

(3) The values of SVI depended on the MLSS concentrations. This dependence was more noticeable with bulked sludges than with those with good settling characteristics.

CONCLUSIONS

1. Complete mixing activated sludge systems encourage excessive growth of filamentous organisms. Aeration systems with the low degree of axial mixing, i.e. with a low dispersion number and higher concentration gradients of substrate through the system cause growth of the non-filamentous microorganisms and suppress the growth of filamentous organisms.

2. It is possible to control the growth of filamentous microorganisms in mixed cultures by maintaining the concentration gradient of the substrate along the aeration system. The concentration gradient of the substrate depends on the hydraulic regime of the aeration system.

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